HOLE OPENER

FIXED-DIAMETER HOLE OPENER

ENLARGEMENT

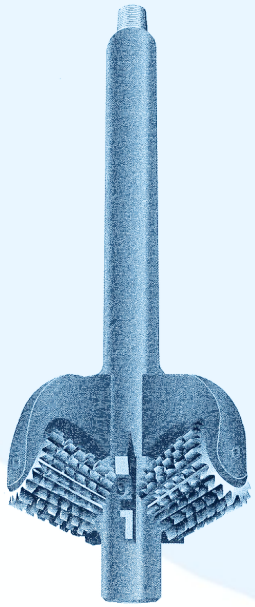
GENERAL DESCRIPTION

FIXED-DIAMETER HOLE OPENER THAT ENLARGES A PREVIOUSLY DRILLED PILOT HOLE FROM THE SURFACE. THE CUTTERS ARE RIGIDLY ATTACHED TO THE BODY: THERE IS NO HYDRAULIC ACTUATION, AND THE TOOL ALWAYS CUTS AT ITS FULL GAUGE. USED MAINLY FOR SURFACE HOLES AND CONDUCTOR CASING, WHERE A BIGGER HOLE GIVES ENOUGH CLEARANCE TO RUN AND CEMENT THE CASING (FOR EXAMPLE, A 17-1/2 IN. HOLE OPENED TO RUN 13-3/8 IN. CASING). THE CHOICE OF CUTTERS LETS IT HANDLE SOFT TO HARD-ABRASIVE FORMATIONS. CUTTER SIZES FROM 26 TO 42 IN.

TECHNICAL CHARACTERISTICS

Std. opening Ø	Min. pilot hole Ø	Fishing neck length	Fishing neck Ø	Top pin API Reg.	Bottom neck length	Bottom neck Ø	Bottom box API Reg.	Body length	Min. bore Ø
26	14	60	9-1/2	6-5/8 - 7-5/8	12	9-1/2	7-5/8	96	1
28	16	60	9-7/8	6-5/8 - 7-5/8	12	9-1/2	7-5/8	96	1
30	18	60	9-7/8	6-5/8 - 7-5/8	12	9-1/2	7-5/8	100	2
32	20	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	100	2
34	22	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	101	2
36	24	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	103	3
38	26	60	10	8-5/8	15	9-1/2	8-5/8	106	3
40	28	60	10	8-5/8	15	9-1/2	8-5/8	106	3
42	30	60	10	8-5/8	15	9-1/2	7-5/8	106	3

Dimensions in inches, approximate. Intermediate sizes or sizes larger than 42 in. available on request.



DESIGN FEATURES

- **Fixed gauge:** cutters rigidly mounted; no hydraulic actuation.
- **Demountable cutters:** easily replaced on the rig floor.
- **Cutters:** sealed-bearing milled tooth or tungsten carbide insert.
- **Long fishing necks:** can be made up below the rotary table despite the large size.
- **Replaceable nozzles:** effective hole cleaning.
- **Wide formation range:** soft to hard-abrasive.

NOZZLES AND CUTTERS

- **Nozzles:** threaded retainer holds the jet nozzle in its sleeve; O-ring seal.
- **Cutter sets:** replacement sets include all parts needed.
- **Rebuild kits:** cutter bearing rebuild kits on special order.

OPERATING PARAMETERS

Formation	Cutter	Weight (lb)	rpm
Soft	Milled tooth	15,000 - 25,000	125 - 150
Medium	Milled tooth	20,000 - 35,000	75 - 85
Hard	Button type	35,000 - 45,000	50 - 60
Extr. hard	TCI button	30,000 - 45,000	30 - 40

Guideline for 19-3/4 - 26 in. holes; smaller sizes need less weight.

GAUGE TOLERANCE (ARMS OPEN)

Size (in)	Plus	Minus
18 - 26	+1/8	- 1/16
27 - 42	+5/32	- 1/16
43 and larger	+3/16	- 1/16

Gauge as dressed, not ground; milled tooth and TCI.

MASTER DRILLER (CONE ARM VERSION)

Cone-arm hole opener, up to 36 in.; one body takes several arm sizes. Bottom box for bit or bullnose. Milled tooth, TCI or PDC cones; soft to medium-hard formations. Body series 8200 to 15000: std. opening 12-1/4 to 36-5/8 in., min. pilot hole 8-1/2 to 24-5/8 in., body Ø 8-1/4 to 15-5/8 in., 640 - 1,900 lb.

HOW TO ORDER

Please specify: 1. pilot hole size 2. hole opening size 3. top and bottom connections if not standard 4. fishing neck diameter if not standard 5. cutter type (milled tooth or tungsten carbide insert) and formation (soft or medium).